

Project Report: Push to Go — F1 Overtake Prediction Simulation

1. Executive Summary

The **Push to Go F1 Simulation** is a real-time data engineering and machine learning pipeline designed to predict Formula 1 overtakes as they happen. By tapping directly into Formula 1's live telemetry WebSockets, the system streams high-frequency data (Speed, Throttle, Brake, RPM), engineers advanced features like Battery State of Charge (SOC), predicts overtakes using an XGBoost model, and visualizes the results on a live Streamlit dashboard.

2. System Architecture & Data Flow

The project is built on a modern, low-latency stack designed for real-time streaming and time-series analysis.

flowchart TD
 %% Define Node Styles
 classDef source fill:#e10600,stroke:#fff,stroke-width:2px,color:#fff,font-weight:bold
 classDef python fill:#ffd43b,stroke:#306998,stroke-width:2px,color:#000,font-weight:bold
 classDef db fill:#419edd,stroke:#fff,stroke-width:2px,color:#fff,font-weight:bold
 classDef dash fill:#ff4b4b,stroke:#fff,stroke-width:2px,color:#fff,font-weight:bold
 %% Nodes
 A((F1 Live Server)):::source
 B[livef1_client.py
Data Ingestion]:::python
 C[SOC Calculator
Feature Engineering]:::python
 D[XGBoost Model
predict.py]:::python
 E[(ScyllaDB / Cassandra
Time-Series DB)]:::db
 F[Streamlit App
app.py]:::dash
 %% Flow
 A -- "SignalR WebSockets\n(CarData.z, TimingData)" --> B
 B -- "Raw Telemetry\n(Speed, Throttle, Brake)" --> C
 C -- "Engineered Features\n(SOC, Gap)" --> D
 D -- "Overtake Prediction" --> E
 B -- "Raw Data" --> E
 E -- "Real-Time Query" --> F

Component Breakdown

1. Data Ingestion (livef1_client.py / fastf1_recorder.py)

- Connects to F1's undocumented SignalR WebSockets.
- Intercepts and decompresses high-frequency streams (CarData.z for telemetry, Position.z for GPS, and TimingData for gaps and sector times).
- Requirement:* A valid F1 TV Access subscription is used to authenticate the connection and unlock the premium telemetry channels.

2. Feature Engineering (src/common/soc_calculator.py)

- Raw telemetry isn't enough to predict an overtake; we need to know how much electrical energy a driver has left.
- The custom **SOC Calculator** estimates the battery's State of Charge by evaluating throttle and braking application over time (e.g., harvesting under braking, deploying under full throttle).

3. Machine Learning Model (`src/ml_model/predict.py`)

- The pipeline passes live Speed, Throttle, Brake, Gap, and SOC into an XGBoost classification model.
- The model outputs a binary prediction indicating whether an overtake is imminent on the current lap.

4. Time-Series Database (`schema.cql`)

- Data is persisted at high frequency into a locally-hosted **ScyllaDB/Cassandra** cluster.
- The keyspace `f1_live.live_telemetry` is clustered by timestamp descending to ensure the dashboard can read the absolute latest data instantly.

5. Real-Time Dashboard (`src/dashboard/app.py`)

- A highly-styled, responsive user interface built with Streamlit and Plotly.
- **Timing Tower:** Shows live driver positions, gaps, tire compounds, and lap counts.
- **Driver Detail:** Interactive gauges for Speed, Throttle, Brake, and RPM.
- **ML Insights:** Displays the real-time "Overtake Likely" prediction alongside the calculated Battery SOC.

3. Data Capture Requirements

*[!IMPORTANT] **Authentication Requirement** F1 enforces strict bandwidth controls on live telemetry. While historical data can be analyzed for free, **live real-time telemetry (CarData.z / Position.z)** requires the client to authenticate with a valid F1 TV Access subscription. Unauthenticated (Guest) clients are heavily throttled and only receive standard TimingData.*

Example Captured Record

When successfully authenticated, the pipeline captures and processes data in the following structure:

```
{
  "driver": "NOR",
  "timestamp": "2026-08-21T15:22:26.837Z",
  "speed": 315.2,
  "throttle": 100.0,
  "brake": 0.0,
  "rpm": 11500,
  "gap_to_ahead": 0.450,
  "estimated_soc": 68.4,
  "overtake_prediction": 1
}
```

4. Current Status & Next Steps

- ☐ **Infrastructure:** ScyllaDB schema and local Docker connectivity are fully configured.
- ☐ **Dashboard:** The Streamlit UI is built, themed (F1 dark mode), and capable of rendering telemetry gauges and speed trap comparisons.
- ☐ **Data Ingestion:** Scripts are ready to intercept SignalR streams.
- ☐ **Action Item:** Ensure an active F1 TV Access subscription is passed into the `fastf1 / livef1` client to unlock the `CarData.z` stream during the next live race session.